

Literature Survey

2.1 Introduction

Infrastructure-as-Code (IaC) has become a fundamental approach for automating cloud infrastructure provisioning by enabling infrastructure resources to be defined through machine-readable configuration files. With the rapid adoption of cloud-native technologies in smart-manufacturing environments, managing secure, scalable, and reliable cloud infrastructure has become increasingly challenging. Recent advances in Large Language Models (LLMs) have created new opportunities for automating Infrastructure-as-Code generation, validation, security analysis, configuration repair, deployment verification, and intelligent decision support. The reviewed studies demonstrate significant progress in these areas while also highlighting several technical limitations that motivate the proposed Telemetry-Aware Multi-Agent Infrastructure-as-Code Assurance System for Smart Manufacturing Cloud Environments.

2.2 Evolution of Research

Early research primarily focused on improving Infrastructure-as-Code generation through retrieval-augmented generation, knowledge injection, and benchmark-based evaluation for Terraform and cloud deployment templates. Later studies introduced security-aware code generation, automated Infrastructure-as-Code repair, Kubernetes misconfiguration detection, enterprise DevSecOps validation, and deployment-oriented evaluation to improve infrastructure reliability and security. More recent research has expanded toward multi-agent reasoning, configuration drift detection, industrial configuration validation, PLC code generation and verification, Industrial Internet of Things (IIoT) security, and intelligent Industrial Control System (ICS) generation. These developments demonstrate the growing role of LLMs in cloud infrastructure and industrial automation while revealing the need for an integrated framework capable of combining validation, deployment verification, runtime telemetry, security assessment, and smart-manufacturing support.

2.3 Literature Review

The reviewed studies collectively demonstrate significant progress in applying Large Language Models (LLMs) to Infrastructure-as-Code (IaC), cloud infrastructure management, and smart-manufacturing systems. Existing research has improved IaC generation through retrieval-augmented generation, knowledge injection, security-aware fine-tuning, automated repair, deployment verification, and multi-agent reasoning. Recent studies have further extended LLM applications to configuration validation, PLC code generation and verification, Industrial IoT security, and Industrial Control System (ICS) automation. While these approaches achieve high accuracy within their respective domains, most focus on a single IaC technology, industrial component, or isolated validation task. Common challenges include limited multi-format support, dependence on static analysis, lack of runtime telemetry integration, limited deployment verification, and minimal coordination between

validation, security assessment, configuration drift detection, and recommendation generation. These limitations motivate the proposed **Telemetry-Aware Multi-Agent Infrastructure-as-Code Assurance System for Smart Manufacturing Cloud Environments**, which integrates these capabilities into a unified intelligent framework.

2.4 Literature Summary

Overall, the reviewed literature demonstrates substantial progress in Infrastructure-as-Code generation, security-aware validation, automated repair, deployment verification, configuration drift detection, industrial configuration validation, PLC verification, and Industrial IoT security using Large Language Models. Despite these advances, most existing approaches address isolated stages of the Infrastructure-as-Code lifecycle or focus on a single technology, such as Terraform, Kubernetes, PLCs, or Industrial Control Systems. A unified framework capable of supporting multiple Infrastructure-as-Code technologies while integrating deployment verification, runtime cloud telemetry, security assessment, configuration drift detection, explainable recommendations, and smart-manufacturing support remains largely unexplored. These recurring research gaps form the foundation for the proposed **Telemetry-Aware Multi-Agent Infrastructure-as-Code Assurance System for Smart Manufacturing Cloud Environments**.